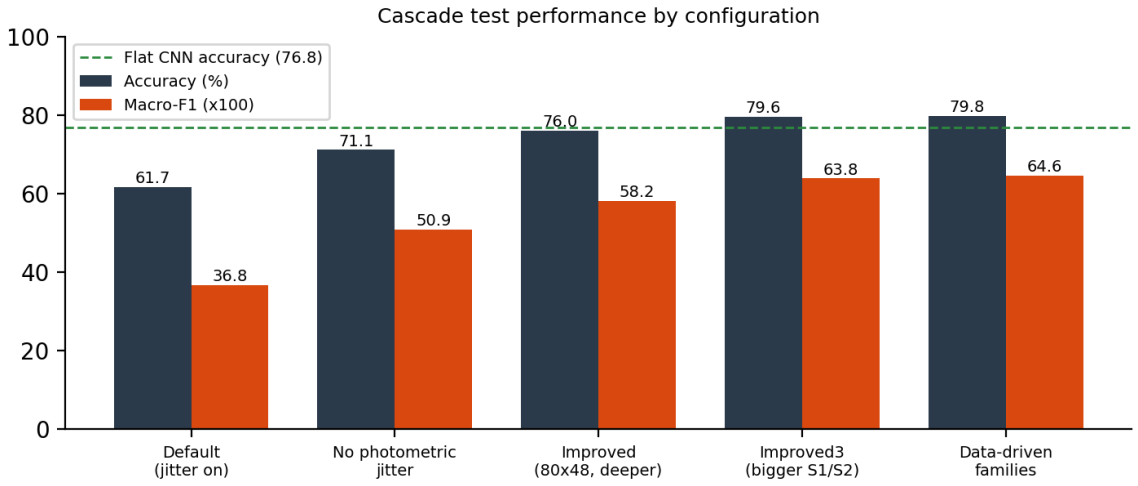
Cascaded-SolarNet: Results, Comparison with Base Papers and Conclusion

Cascaded Convolutional Neural Network for Solar Panel Defect Classification (InfraredSolarModules, 12 classes). All numbers are measured on the same held-out test split of 3,000 images (stratified 70/15/15, seed 42), single run per configuration. Code, configs and raw metrics: github.com/techwack/cascaded-cnn.

1. Results

The best cascade reaches **79.8%** accuracy, macro-F1 **0.646** and Cohen's kappa **0.717** on the 12-class problem (data-driven families), and **94.1%** accuracy on the Stage 1 healthy-versus-defect decision. With the semantic families (Electrical, Environmental, Structural) the 'improved3' configuration reaches 79.6% accuracy and macro-F1 0.638.

Configuration	Model	Accuracy	Macro-F1	Kappa	Stage 1 binary acc.
Default (brightness/contrast/noise jitter)	Cascade	61.7%	0.368	0.461	81.0%
	Flat CNN (same split)	47.6%	0.379	0.357	n/a
No photometric jitter (flips only)	Cascade	71.1%	0.509	0.583	86.0%
	Flat CNN (same split)	68.5%	0.571	0.586	n/a
Improved: 80x48 input, 2 convs/block	Cascade	76.0%	0.582	0.656	89.5%
	Flat CNN (same split)	76.8%	0.653	0.689	n/a
Improved2: + balanced sampling, label smoothing	Cascade	75.5%	0.576	0.648	89.1%
Improved3: + larger Stage 1/2 CNNs	Cascade	79.6%	0.638	0.714	93.8%
Data-driven family grouping	Cascade	79.8%	0.646	0.717	94.1%



Main findings

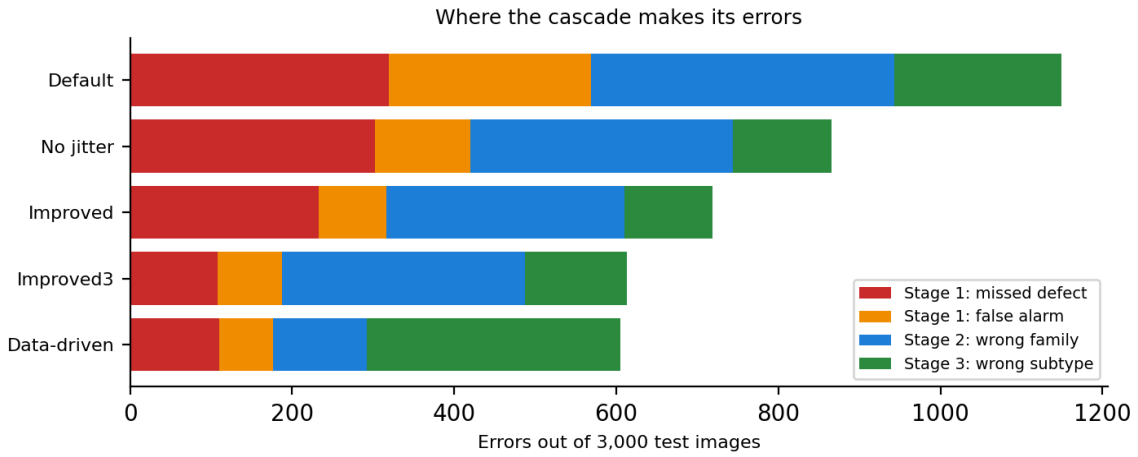
- **Removing photometric jitter was the biggest single gain.** Brightness, contrast and noise augmentation lowered cascade accuracy to 61.7%; flips only gave 71.1%. In thermal images brightness is temperature, so jittering it destroys the signal that separates the classes.

- **Larger input and capacity helped next.** Upsampling to 80x48 and using two convolutions per block raised accuracy to 76.0%; enlarging the Stage 1 and Stage 2 networks raised it to 79.6%, and Stage 1 binary accuracy from 89.5% to 93.8%.

- **Cascade versus flat CNN.** The cascade is ahead on accuracy (79.6% vs 76.8%) and kappa (0.714 vs 0.689) but slightly behind on macro-F1 (0.638 vs 0.653), so the advantage is modest. 51.0% of test images leave the cascade at Stage 1 and never reach the larger networks.

- **Error attribution.** With the semantic families, wrong-family decisions at Stage 2 are the largest error source. The data-driven grouping cut Stage 2 errors from 300 to 116, but the hard confusions moved into Stage 3 (126 to 313 errors), so the net gain is small (79.6% to 79.8%, within single-run noise).

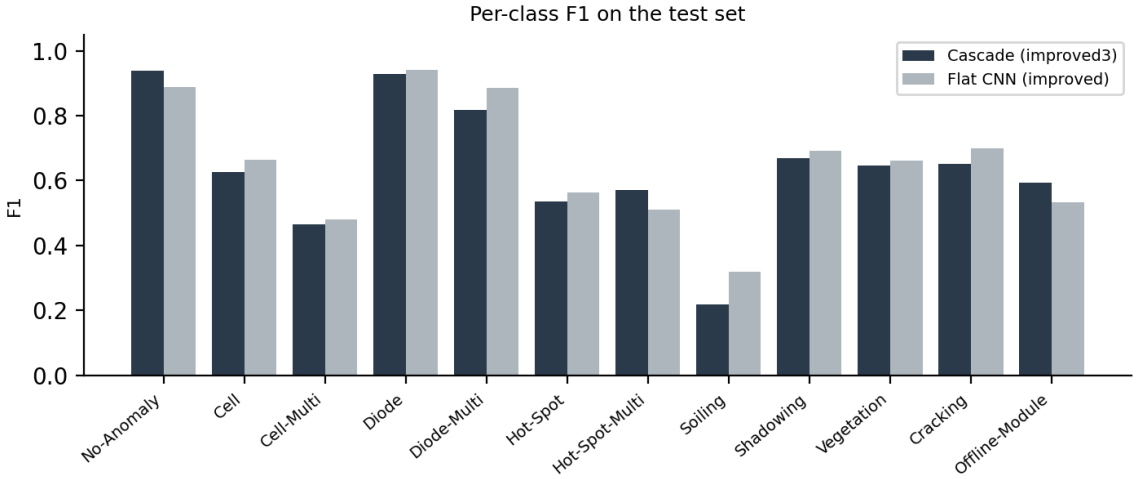
- **Tried without gain:** soft routing (multiplying stage probabilities), flip test-time augmentation, tuning the Stage 1 threshold on validation data (optimum stayed at 0.5) and balanced sampling with label smoothing.



Per-class performance

Rare classes are the weak point: they have only 123 to 250 training images, and several look alike (Cell vs Cell-Multi, Hot-Spot vs Cell). Class weighting in the loss, stratified splitting and macro-F1 were used to keep them from being ignored, but they remain the hardest classes.

Class	Test images	Cascade F1 (improved3)	Cascade F1 (data-driven)	Flat CNN F1
No-Anomaly	1500	0.94	0.94	0.89
Cell	281	0.63	0.64	0.66
Cell-Multi	193	0.47	0.44	0.48
Diode	225	0.93	0.92	0.94
Diode-Multi	26	0.82	0.91	0.88
Hot-Spot	37	0.54	0.40	0.56
Hot-Spot-Multi	37	0.57	0.58	0.51
Soiling	31	0.22	0.30	0.32
Shadowing	159	0.67	0.68	0.69
Vegetation	246	0.65	0.63	0.66
Cracking	141	0.65	0.66	0.70
Offline-Module	124	0.59	0.65	0.53



2. Comparison with the Base Papers

The base work in our literature survey falls into two groups: results on the same InfraredSolarModules benchmark, and results on other data, modalities or tasks. Only the first group is a fair comparison, and even there the task (binary or 12-class) is not stated in the survey. We therefore compare our Stage 1 binary accuracy and our 12-class accuracy separately.

Source (as cited in our survey)	Data / task	Reported	Our result	Comparable?
Millendorf et al. 2020 (dataset): Random Forest	InfraredSolarModules, task not stated	84%	Binary 94.1%; 12-class 79.8%	Same data. Task unclear.
Millendorf et al. 2020: custom CNN	InfraredSolarModules, task not stated	92.5%	Binary 94.1%; 12-class 79.8%	Same data. Task unclear.
Millendorf et al. 2020: EfficientNet + SVM	InfraredSolarModules, task not stated	93%	Binary 94.1%; 12-class 79.8%	Same data. Task unclear.
PV-faultNet (arXiv:2411.02997)	Lightweight CNN, dataset not specified in survey	93% at 8 ms/image	Not measured on same hardware	No (different setup)

Source (as cited in our survey)	Data / task	Reported	Our result	Comparable?
Tella et al. 2025 (8 CNNs/ViTs)	ELPV benchmark (different dataset)	about 68% multi-class, 73% binary	12-class 79.8%	Directional only
B-CNN (Pathak and Patil 2025)	Thermal + visual fusion (two sensors)	98.5% accuracy	Single IR sensor	No (more information)
Field-adapted ResNet-50	Field data, transfer learning	98.59%	-	No (different data)
CNN + BFO (VGG-16)	Pretrained CNNs on other data	98.75%	-	No (different data)

How to read this.

- **Binary (healthy vs defect):** our Stage 1 reaches 94.1%, which is at or above the 92.5% to 93% figures if those refer to the binary task, as their magnitude suggests. The margin is 1 to 2 points on a single run and a different protocol, so we claim parity rather than a clear win.

- **12-class:** our 79.8% is not comparable to 93%. If the 92.5% to 93% figures were 12-class results, our model would be about 13 points below them. It is, however, above the roughly 68% multi-class level that Tella et al. report across eight architectures, although on a different dataset.

- **Models above 98%** use two sensors, pretrained backbones on other data or field-specific data. They show what additional information can buy, not a like-for-like baseline for a single-camera, 24x40 pixel IR model.

- **Our own baselines:** the flat CNN with identical building blocks reaches 76.8% accuracy and 0.653 macro-F1. A random forest baseline was planned but not run on our final split.

Note on sources. The original dataset paper (Millendorf et al., ICLR 2020 workshop, 5 pages) describes the data and lists benchmark classification results as a goal, but it reports no model accuracies. The 84%, 92.5% and 93% figures in our survey are quoted from elsewhere and could not be traced to that paper. They should be verified against their original source before being presented as published baselines. The report's stated target of above 95% accuracy was not reached.

3. Conclusion

- We implemented Cascaded-SolarNet, a three-stage cascade of five small CNNs for 12-class infrared solar module defect classification, in PyTorch, with a reproducible split, tests, diagrams and live demos.

- **The cascade works.** After diagnosing and removing harmful photometric augmentation, enlarging the input and strengthening the front stages, accuracy rose from 61.7% to 79.8% (macro-F1 0.37 to 0.65).
Healthy-versus-defect triage reaches about 94.1%, in line with published binary results on this dataset.

- **The claim that a cascade clearly beats a flat classifier was only partly supported.** It is ahead on accuracy and kappa and processes about half of all images with only the small Stage 1 network, but a flat CNN with the same building blocks is within about 3 accuracy points and slightly better on macro-F1. The target of exceeding 93% on all 12 classes was not met.

- **Error propagation is the central trade-off.** A wrong decision at Stage 1 or Stage 2 cannot be undone.
Re-grouping the classes by their actual confusions moved errors between stages rather than removing them, and soft routing did not recover them.

- **Thermal images need thermal-aware augmentation.** Geometric flips help; brightness, contrast and noise jitter hurt, because brightness encodes temperature.

Limitations

- Single run per configuration (seed 42); differences below about 0.5 points are within noise and no confidence intervals were computed.

- The benchmark has no module or site identifiers, so near-duplicate crops can appear in both training and test data, which may inflate all reported numbers, ours and published ones.

- Rare classes (Soiling, Hot-Spot, Diode-Multi) remain weak, which keeps macro-F1 well below accuracy.

- Latency on edge hardware was not measured reliably; the early-exit rate (about half of the images) is the supporting evidence for efficiency.

- The comparison with published results is limited by missing information about which task the quoted baselines address.

Future work

- Run several seeds and report mean and standard deviation; run the random forest and an EfficientNet baseline on the same split.

- Improve Stage 2, for example by training the family classifier jointly with the specialists or on the cascade's own routing errors, or with class-specific augmentation for rare faults.

- Add the planned ablations (no early exit, unweighted loss, upsampling factor in isolation) and deduplicate near-identical crops before splitting.

- Measure real latency and memory on a drone-class edge device, and test additional modalities such as RGB.